

Chapter 6 Test model

Multiple Choice Questions (70)

1. Which of the following best describes the origin of the term “SQL”?
 - a. It stands for Simple Query Language
 - b. It was derived from “Structured Query Language” from the start
 - c. It originated from “SEQUEL,” which stood for “SEQUEL TO SQUARE”
 - d. It was named after a database pioneer named Sequel

2. In SQL, which clause is used to enforce that a foreign key value must either match a primary key in another table or be NULL?
 - a. PRIMARY KEY
 - b. UNIQUE
 - c. FOREIGN KEY
 - d. CHECK

3. What is the correct SQL statement to define a domain for Social Security Numbers as a 9-character string?
 - a. CREATE TYPE SSN_TYPE AS CHAR(9);
 - b. CREATE DOMAIN SSN_TYPE AS CHAR(9);
 - c. DEFINE DOMAIN SSN_TYPE CHAR(9);
 - d. DECLARE SSN_TYPE CHAR(9);

4. Which of the following is NOT a valid SQL data type according to the standard covered in Chapter 6?
 - a. BIT VARYING(n)
 - b. TIMESTAMP WITH TIME ZONE
 - c. JSON_OBJECT
 - d. DOUBLE PRECISION

5. In the SQL CREATE TABLE statement, what does the clause “ON DELETE SET NULL” do?

- a. Deletes the referenced row and all dependent rows
 - b. Sets the foreign key value to NULL when the referenced row is deleted
 - c. Prevents deletion if dependent rows exist
 - d. Replaces the foreign key with a default value
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6. Which SQL clause eliminates duplicate rows from the result set?

- a. UNIQUE
 - b. DISTINCT
 - c. NO DUPLICATES
 - d. GROUP BY
-

7. What is the result of a SELECT statement with no WHERE clause and two tables in the FROM clause?

- a. An inner join
 - b. A natural join
 - c. A Cartesian product
 - d. An error due to ambiguity
-

8. Which of the following is true about the BOOLEAN data type in SQL?

- a. It only accepts TRUE or FALSE
 - b. It accepts TRUE, FALSE, or NULL
 - c. It is not part of the SQL standard
 - d. It is stored as 1 or 0 internally
-

9. In SQL, what does the pattern '18901' match using the LIKE operator?

- a. Any SSN starting with 1 and ending with 8901
 - b. Any 9-character string where the 3rd character is '1' and the last four are '8901'
 - c. Any string containing '18901'
 - d. Any SSN with exactly two digits before '1' and three between '1' and '8901'
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10. Which of the following correctly uses tuple variables (aliases) to find employees and their supervisors?

- a. SELECT Fname, Lname FROM EMPLOYEE E, EMPLOYEE S WHERE E.Super_ssn = S.Ssn;
- b. SELECT E.Fname, E.Lname, S.Fname, S.Lname FROM EMPLOYEE AS E,

EMPLOYEE AS S WHERE E.Super_ssn = S.Ssn;

c. SELECT Fname, Lname FROM EMPLOYEE JOIN EMPLOYEE ON Super_ssn = Ssn;

d. SELECT * FROM EMPLOYEE WHERE Super_ssn IN (SELECT Ssn FROM EMPLOYEE);

11. What is the purpose of the CHECK clause at the table level (not attribute level)?

- a. To enforce referential integrity across tables
 - b. To ensure that a condition involving multiple attributes holds for every tuple
 - c. To validate input against a regular expression
 - d. To restrict the number of rows in a table
-

12. Which of the following is a valid use of the DEFAULT clause in SQL?

- a. Dno INT DEFAULT 5 CHECK (Dno > 0);
 - b. Salary FLOAT DEFAULT NULL NOT NULL;
 - c. Name VARCHAR(30) DEFAULT 'Unknown' PRIMARY KEY;
 - d. Ssn CHAR(9) DEFAULT UNIQUE;
-

13. What does the SQL standard SQL-1999 introduce that earlier versions did not?

- a. Support for XML
 - b. Core specification plus specialized extensions
 - c. Full object-relational features
 - d. The SELECT-FROM-WHERE syntax
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14. In SQL, what is the difference between CHAR(n) and VARCHAR(n)?

- a. CHAR(n) stores exactly n characters, padding with spaces if needed; VARCHAR(n) stores up to n characters without padding
 - b. CHAR(n) is case-sensitive; VARCHAR(n) is not
 - c. VARCHAR(n) is faster for indexing
 - d. CHAR(n) cannot be used in primary keys
-

15. Which SQL command is used to create a virtual relation that does not correspond to a physical file?

- a. CREATE TABLE
- b. CREATE SCHEMA

- c. CREATE VIEW
 - d. CREATE DOMAIN
-

16. What is the correct way to specify a composite primary key in SQL?
- a. PRIMARY KEY (A, B)
 - b. PRIMARY KEY A, B
 - c. KEY (A) AND KEY (B)
 - d. UNIQUE (A, B) PRIMARY
-

17. Which of the following is NOT enforced by the entity integrity constraint?
- a. Primary key attributes cannot be NULL
 - b. Primary key values must be unique
 - c. Foreign key values must reference existing primary keys
 - d. All of the above are enforced
-

18. In the COMPANY schema, which referential action is most appropriate for the WORKS_ON table when an EMPLOYEE is deleted?
- a. ON DELETE CASCADE
 - b. ON DELETE SET NULL
 - c. ON DELETE RESTRICT
 - d. ON DELETE SET DEFAULT
-

19. What is the effect of using "UNION ALL" instead of "UNION" in SQL?
- a. It removes duplicates
 - b. It preserves duplicates
 - c. It sorts the result
 - d. It performs intersection
-

20. Which of the following is true about SQL's treatment of relations compared to the relational model?
- a. SQL enforces set semantics strictly
 - b. SQL allows duplicate tuples (multiset semantics)
 - c. SQL requires tuple IDs for all tables
 - d. SQL does not support NULL values

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21. How can you retrieve all attributes from the EMPLOYEE table using the SELECT clause?
- a. SELECT ALL FROM EMPLOYEE;
 - b. SELECT *FROM EMPLOYEE*;
 - c. *SELECT EMPLOYEE.* FROM EMPLOYEE;
 - d. Both b and c
-

22. What is the purpose of the CONSTRAINT keyword in SQL?
- a. To define a new data type
 - b. To name a constraint for future reference (e.g., ALTER or DROP)
 - c. To enforce NOT NULL automatically
 - d. To create a foreign key without referencing a table
-

23. Which of the following is a valid INTERVAL data type usage?
- a. INTERVAL '5' YEAR
 - b. INTERVAL DATE '2020-01-01'
 - c. INTERVAL BETWEEN 10 AND 20
 - d. INTERVAL (DAY TO SECOND)
-

24. In SQL, what does the BETWEEN operator include?
- a. Only the lower bound
 - b. Only the upper bound
 - c. Both bounds inclusively
 - d. Neither bound
-

25. Which clause is used to sort query results in descending order?
- a. SORT BY DESC
 - b. ORDER DESC
 - c. ORDER BY ... DESC
 - d. GROUP BY ... DESC
-

26. What is the correct syntax to give all employees in department 5 a 10% salary raise?
- a. UPDATE EMPLOYEE SET Salary = Salary 1.1 *WHERE Dno = 5*;

- b. *UPDATE EMPLOYEE SET Salary = 1.1 WHERE Dno = 5;*
 - c. *MODIFY EMPLOYEE Salary = Salary 1.1 WHERE Dno = 5;*
 - d. *SET EMPLOYEE.Salary = Salary * 1.1 FOR Dno = 5;*
-

27. Which SQL standard first introduced XML features?

- a. SQL-1999
 - b. SQL-2003
 - c. SQL-2006
 - d. SQL-2008
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28. What is the maximum number of positions in the standard SQL DATE type?

- a. 8
 - b. 10
 - c. 12
 - d. Variable
-

29. Which of the following is true about the CASCADE option in referential integrity?

- a. It is suitable for “relationship” relations like WORKS_ON
 - b. It sets foreign keys to NULL on deletion
 - c. It prevents any deletion that would violate integrity
 - d. It only works with primary keys
-

30. What happens if you omit the WHERE clause in a DELETE statement?

- a. No rows are deleted
 - b. Only the first row is deleted
 - c. All rows in the table are deleted
 - d. An error is thrown
-

31. Which of the following correctly uses arithmetic in a SELECT clause to compute a 10% raise?

- a. *SELECT Fname, Lname, Salary + 0.1 AS NewSal ...*
- b. *SELECT Fname, Lname, Salary * 1.1 AS Increased_sal ...*
- c. *SELECT Fname, Lname, 10% Salary AS NewSal ...*
- d. *SELECT Fname, Lname, RAISE(Salary, 10%) ...*

32. In SQL, what is a “catalog”?

- a. A single table
 - b. A collection of schemas
 - c. A type of constraint
 - d. A backup of the database
-

33. Which of the following is NOT a basic constraint type in the relational model supported by SQL?

- a. Key constraint
 - b. Entity integrity
 - c. Domain constraint
 - d. Referential integrity
-

34. What is the purpose of the AS keyword in `SELECT E.Fname AS FirstName ...`?

- a. To create a new column in the table
 - b. To rename the output column in the result
 - c. To define a view
 - d. To alias the table name
-

35. Which operator matches any single character in a LIKE pattern?

- a. %
 - b. *
 - c. _
 - d. ?
-

36. What is the result of: `SELECT * FROM EMPLOYEE, DEPARTMENT;` with no WHERE clause?

- a. Natural join
 - b. Equi-join
 - c. Cartesian product
 - d. Empty result
-

37. Which of the following is a valid use of the CREATE SCHEMA statement?

- a. CREATE SCHEMA COMPANY AUTHORIZATION 'Jsmith';
 - b. CREATE SCHEMA TABLE EMPLOYEE (...);
 - c. CREATE SCHEMA VIEW V1 AS ...;
 - d. CREATE SCHEMA DOMAIN D1 AS INT;
-

38. In SQL, what does “WITH DATA” do in a CREATE TABLE ... LIKE statement?

- a. Creates only the structure
 - b. Copies both structure and data
 - c. Encrypts the data
 - d. Compresses the table
-

39. Which data type would you use to store a precise timestamp with time zone information?

- a. DATE
 - b. TIME
 - c. TIMESTAMP WITH TIME ZONE
 - d. INTERVAL
-

40. What is the correct way to enforce that Dnumber must be between 1 and 20?

- a. Dnumber INT CHECK (Dnumber > 0 AND Dnumber < 21);
 - b. Dnumber INT BETWEEN 1 AND 20;
 - c. Dnumber INT CONSTRAINT range CHECK (1–20);
 - d. Dnumber INT VALID (1,20);
-

41. Which of the following statements about NULL in SQL is true?

- a. NULL = NULL evaluates to TRUE
 - b. NULL is the same as zero or empty string
 - c. NULL represents unknown or inapplicable value
 - d. NULL cannot appear in foreign keys
-

42. What is the purpose of the EXCEPT operator in SQL?

- a. Returns rows in the first query but not in the second
- b. Returns common rows between two queries
- c. Combines all rows from two queries
- d. Subtracts numeric values

43. Which of the following is true about user-defined types (UDTs) in SQL?

- a. Created using CREATE DOMAIN
 - b. Supported for object-oriented applications via CREATE TYPE
 - c. Cannot be used in table definitions
 - d. Replaced by VARCHAR in modern SQL
-

44. In the COMPANY schema, what is the likely referential action for DEPARTMENT.Mgr_ssn referencing EMPLOYEE.Ssn?

- a. ON DELETE CASCADE
 - b. ON DELETE SET NULL
 - c. ON DELETE RESTRICT
 - d. ON DELETE NO ACTION
-

45. What does the query "SELECT DISTINCT Dno FROM EMPLOYEE;" return?

- a. All department numbers including duplicates
 - b. Only unique department numbers
 - c. Employee names grouped by department
 - d. An error because Dno is not a key
-

46. Which clause is mandatory in a basic SQL retrieval query?

- a. WHERE
 - b. FROM
 - c. ORDER BY
 - d. GROUP BY
-

47. What is the correct interpretation of:

UPDATE EMPLOYEE SET SALARY = SALARY * 1.1 WHERE DNO IN (SELECT DNUMBER FROM DEPARTMENT WHERE DNAME='Research');

- a. Only one employee gets a raise
 - b. All employees in Research department get a 10% raise
 - c. All departments get a budget increase
 - d. Salaries are reset to 1.1
-

48. Which of the following is NOT a numeric data type in standard SQL?

- a. INTEGER
 - b. FLOAT
 - c. DECIMAL
 - d. All are valid
-

49. What is the primary purpose of the SQL schema concept?

- a. To store backup copies
 - b. To group related tables, views, and constraints under an authorization identifier
 - c. To define user passwords
 - d. To compress data
-

50. Which of the following queries correctly uses aliasing to avoid ambiguity?

- a. SELECT Fname FROM EMPLOYEE E, DEPARTMENT D WHERE E.Dno = D.Dnumber;
 - b. SELECT E.Fname FROM EMPLOYEE E, DEPARTMENT D WHERE E.Dno = D.Dnumber;
 - c. SELECT Fname FROM EMPLOYEE, DEPARTMENT WHERE Dno = Dnumber;
 - d. Both a and b
-

51. What is the minimum number of characters represented by '%' in a LIKE pattern?

- a. 0
 - b. 1
 - c. 2
 - d. Undefined
-

52. Which SQL command is used to remove a table and its data permanently?

- a. DELETE TABLE
 - b. DROP TABLE
 - c. REMOVE TABLE
 - d. ERASE TABLE
-

53. In SQL, what is the default behavior when a referential integrity violation occurs during an UPDATE?

- a. CASCADE

- b. SET NULL
 - c. Reject the update
 - d. Ignore the constraint
-

54. Which of the following is a valid table-level CHECK constraint?
- a. CHECK (Salary > 0)
 - b. CHECK (Dno > 0 AND Dno < 21)
 - c. CHECK (Dept_create_date <= Mgr_start_date)
 - d. All of the above
-

55. What is the result of: SELECT 100 / 3; in SQL?
- a. 33
 - b. 33.333...
 - c. Depends on data types (integer vs float)
 - d. Error
-

56. Which of the following is true about the INTERSECT operator?
- a. Returns rows that appear in both queries
 - b. Requires DISTINCT to work
 - c. Is not part of SQL standard
 - d. Only works on numeric data
-

57. What does the statement "CREATE TABLE D5EMPS LIKE EMPLOYEE ... WITH DATA;" do?
- a. Creates an empty table with same structure
 - b. Creates a table with same structure and copies data for Dno=5
 - c. Creates a view
 - d. Creates a domain
-

58. Which of the following is NOT part of the basic SELECT-FROM-WHERE structure?
- a. Projection attributes
 - b. Selection condition
 - c. Join condition (when multiple tables)
 - d. Transaction control

59. In SQL, how are attribute names qualified to resolve ambiguity?

- a. Using dot notation: RelationName.AttributeName
- b. Using underscore: RelationName_AttributeName
- c. Using parentheses
- d. Using AS only

60. What is the correct way to insert a new employee with partial attributes specified?

- a. INSERT INTO EMPLOYEE VALUES ('John', 'B', 'Smith', ...);
- b. INSERT INTO EMPLOYEE (Fname, Lname, Ssn) VALUES ('John', 'Smith', '123456789');
- c. ADD ROW TO EMPLOYEE ...
- d. Both a and b are valid

61. Which of the following is a very hard question: What is the theoretical foundation of SQL?

- a. Relational algebra
- b. Tuple relational calculus
- c. Entity-relationship model
- d. Object-oriented model

62. Which SQL feature allows a new table to be created from a query result?

- a. CREATE TABLE ... AS SELECT ...
- b. INSERT INTO ... SELECT ...
- c. CREATE VIEW
- d. Both a and b

63. What is the main difference between SQL-92 and SQL-1999?

- a. SQL-1999 introduced core + extensions model
 - b. SQL-92 had no JOIN syntax
 - c. SQL-1999 removed NULLs
 - d. SQL-92 supported XML
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64. Which of the following constraints can reference multiple attributes in a single CHECK clause?
- a. Attribute-level CHECK
 - b. Table-level CHECK
 - c. PRIMARY KEY
 - d. FOREIGN KEY
-

65. In the absence of an ORDER BY clause, what determines the order of rows in the result?
- a. Primary key order
 - b. Insertion order
 - c. Undefined (implementation-dependent)
 - d. Alphabetical by first column
-

66. Which of the following is true about the TIMESTAMP data type?
- a. It includes date, time, and at least six fractional second digits
 - b. It cannot include time zone
 - c. It is the same as DATE
 - d. It is deprecated in SQL-2008
-

67. What is the purpose of the "AUTHORIZATION" clause in CREATE SCHEMA?
- a. To specify the password
 - b. To identify the schema owner
 - c. To grant permissions
 - d. To encrypt the schema
-

68. Which of the following is a valid reason to use domains in SQL?
- a. To simplify changing data types across many attributes
 - b. To improve query performance
 - c. To replace foreign keys
 - d. To enforce encryption
-

69. In a very hard scenario: Which SQL construct most closely corresponds to the relational algebra selection (σ)?
- a. SELECT clause
 - b. FROM clause

- c. WHERE clause
 - d. GROUP BY clause
-

70. Which of the following best describes SQL's deviation from the pure relational model?
- a. It enforces strict set semantics
 - b. It allows duplicate tuples (bag semantics)
 - c. It prohibits NULLs
 - d. It requires all tables to have tuple IDs
-

Answer Key – MCQs (1–70)

- 1. c
- 2. c
- 3. b
- 4. c
- 5. b
- 6. b
- 7. c
- 8. b
- 9. d
- 10. b
- 11. b
- 12. a
- 13. b
- 14. a
- 15. c
- 16. a
- 17. c
- 18. a
- 19. b
- 20. b
- 21. d
- 22. b
- 23. a
- 24. c
- 25. c
- 26. a

27. c

28. b

29. a

30. c

31. b

32. b

33. c

34. b

35. c

36. c

37. a

38. b

39. c

40. a

41. c

42. a

43. b

44. c

45. b

46. b

47. b

48. d

49. b

50. d

51. a

52. b

53. c

54. c

55. c

56. a

57. b

58. d

59. a

60. d

61. b

62. a

63. a

64. b

65. c

- 66. a
 - 67. b
 - 68. a
 - 69. c
 - 70. b
-

True/False Questions (30)

1. SQL was originally called SEQUEL, which stood for “Structured English Query Language.”
 - A. True
 - B. False

2. In SQL, the relational model’s set semantics are strictly enforced, and duplicate tuples are automatically removed.
 - A. True
 - B. False

3. The DEFAULT clause can be used together with NOT NULL to provide a fallback value.
 - A. True
 - B. False

4. A FOREIGN KEY constraint can reference a UNIQUE key, not just a PRIMARY KEY.
 - A. True
 - B. False

5. The CHECK constraint at the table level can involve multiple attributes from the same tuple.
 - A. True
 - B. False

6. SQL-92 is also known as SQL-2.
 - A. True

B. False

7. The CREATE DOMAIN statement improves schema maintainability by centralizing data type definitions.

A. True

B. False

8. In SQL, "%" in a LIKE pattern matches exactly one character.

A. True

B. False

9. The BETWEEN operator is inclusive of both boundary values.

A. True

B. False

10. The CASCADE option for referential integrity is generally unsafe for entity tables but suitable for relationship tables.

A. True

B. False

11. The BOOLEAN data type in SQL cannot store NULL.

A. True

B. False

12. A missing WHERE clause in a DELETE statement deletes the entire table structure.

A. True

B. False

13. The INTERSECT operator returns only rows that appear in both result sets.

A. True

B. False

14. SQL allows two tables in the same schema to have identical names.

- A. True
 - B. False
-

15. The TIMESTAMP data type includes fractional seconds by default.

- A. True
 - B. False
-

16. In SQL, attribute names must be unique within a table but can be reused across tables.

- A. True
 - B. False
-

17. The AS keyword is mandatory when creating table aliases.

- A. True
 - B. False
-

18. The UNION operator removes duplicates by default.

- A. True
 - B. False
-

19. SQL-3 refers to the standard that began with SQL-1999.

- A. True
 - B. False
-

20. The entity integrity constraint allows primary key attributes to be NULL if the table is empty.

- A. True
 - B. False
-

21. The ORDER BY clause can sort by column position (e.g., ORDER BY 1, 2).

- A. True
 - B. False
-

22. The INSERT statement can insert multiple rows using a subquery.

- A. True
 - B. False
-

23. A schema in SQL is a named collection of tables, views, and constraints under an authorization identifier.

- A. True
 - B. False
-

24. The CHAR(n) type always uses exactly n bytes of storage, regardless of actual content length.

- A. True
 - B. False
-

25. The EXCEPT operator is equivalent to the relational algebra difference operation.

- A. True
 - B. False
-

26. In SQL, "NULL = NULL" evaluates to TRUE.

- A. True
 - B. False
-

27. The UPDATE statement can modify multiple attributes in a single command.

- A. True
 - B. False
-

28. The SQL standard fully supports object-oriented features as of SQL-1999.

- A. True
 - B. False
-

29. A catalog in SQL contains one or more schemas.

- A. True
 - B. False
-

30. The LIKE operator is case-sensitive in all SQL implementations.

- A. True
 - B. False
-

Answer Key – True/False (1–30)

- 1. B (False – SEQUEL stood for “SEQUEL TO SQUARE”)
- 2. B
- 3. A
- 4. A
- 5. A
- 6. A
- 7. A
- 8. B
- 9. A
- 10. A
- 11. B
- 12. B (deletes rows, not structure)
- 13. A
- 14. B
- 15. A
- 16. A
- 17. B
- 18. A
- 19. A
- 20. B
- 21. A
- 22. A

- 23. A
 - 24. A
 - 25. A
 - 26. B
 - 27. A
 - 28. B (OO features added in SQL-2008)
 - 29. A
 - 30. B (case sensitivity is implementation-dependent)
-